Lab 8
Getting Started with Analog to Digital Conversion and Sampling

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I. Introduction

This lab had two main objectives. The first was to set up a DAQ board, which acts as an analog to digital converter (ADC), and control it from MATLAB to bring a signal from the function generator onto the computer. The second was to explore the two main limits of this process, the voltage resolution of the ADC and its sampling rate, by acquiring sine waves under different settings and recording audio at several sampling rates.

II. Results

1. Setting Up the NI USB-DAQ

Two wires were connected to the first analog input terminals of the DAQ, AI 0+ and AI 0−, and the DAQ was plugged into the computer. In MATLAB, daq.getDevices found the board as Dev5, with an input range of −10 to +10 V and sampling rates up to 20,000 samples/s. A DAQ session was then created, and channel ai0 was added as a voltage input with addAnalogInputChannel(s, 'Dev5', 'ai0', 'Voltage').

2. Acquiring a Test Signal

The function generator was connected to both the DAQ and the oscilloscope and set to a 500 Hz sine wave with a 2 V amplitude (4 Vpp) and no DC offset. The DAQ was set to Fs = 10,000 samples/s and 100 samples, so each recording lasted:

$$T = 100/10{,}000 = 0.01 \text{ s}$$

Since the period of the signal is 2 ms, this should capture 5 cycles of 20 samples each. The plot (Fig. 1) showed exactly that, with peaks at about ±2.0 V, matching the 4 Vpp on the oscilloscope.

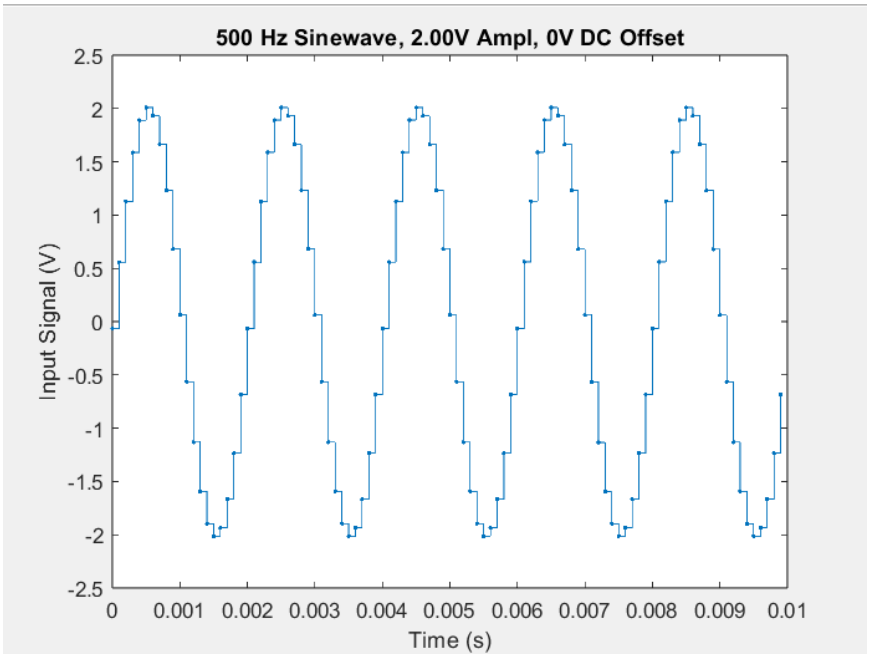


Figure 1: 500 Hz, 2 V amplitude sine wave acquired at 10,000 samples/s.

3. Configuration 1: Low Sampling Rate

The function generator was set to a 200 Hz sine wave with a 200 mV amplitude (400 mVpp), and the DAQ was set to 1500 samples/s for 0.01 s. This gave only 15 samples over 2 cycles, or:

Fs/f = 1500/200 = 7.5 samples per period

The plot (Fig. 2) did not properly represent the sine wave. It showed a blocky staircase, and the two cycles did not match: around the first peak the samples were 0.197 V then 0.164 V, and around the second they were 0.160 V then 0.198 V.

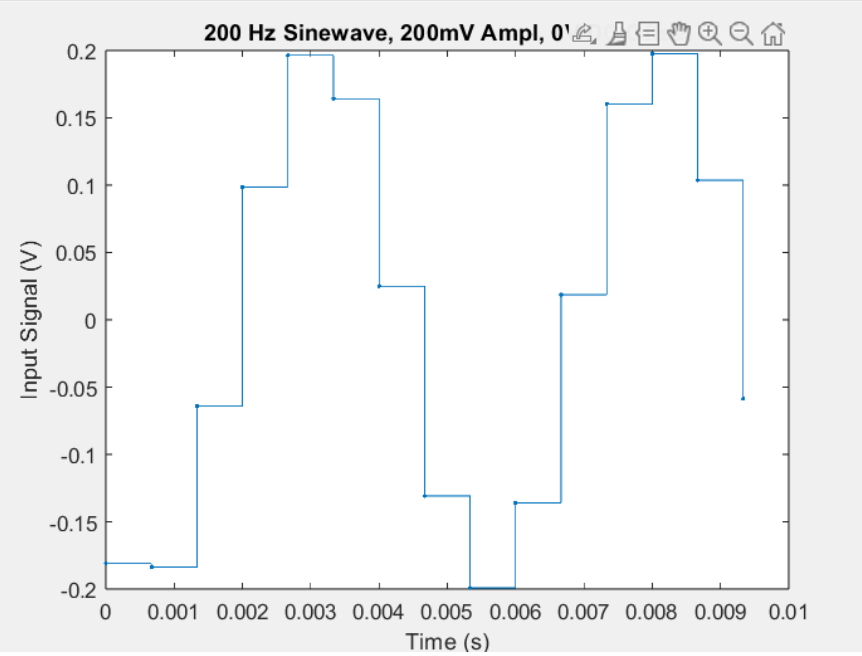


Figure 2: 200 Hz, 200 mV amplitude sine wave acquired at 1500 samples/s (Configuration 1).

The cause was sampling too slowly. With only 7.5 samples per period, the samples landed at different points on each cycle, so the shape could not be traced. Since 1500 samples/s is still more than twice 200 Hz, the frequency was captured correctly, just not the shape. This could have been avoided by raising the sampling rate, for example to 10,000 samples/s for 50 samples per period.

4. Configuration 2: Small Signal Amplitude

The amplitude was lowered to 10 mV (20 mVpp) at 200 Hz, and the sampling rate was raised to 10,000 samples/s for 0.01 s, or 50 samples per period. The plot (Fig. 3) showed the correct frequency, but the signal appeared as a staircase of levels about 1.3 mV apart, and the peaks at 11.0 mV and troughs at −8.4 mV were flat for 5 to 7 samples in a row.

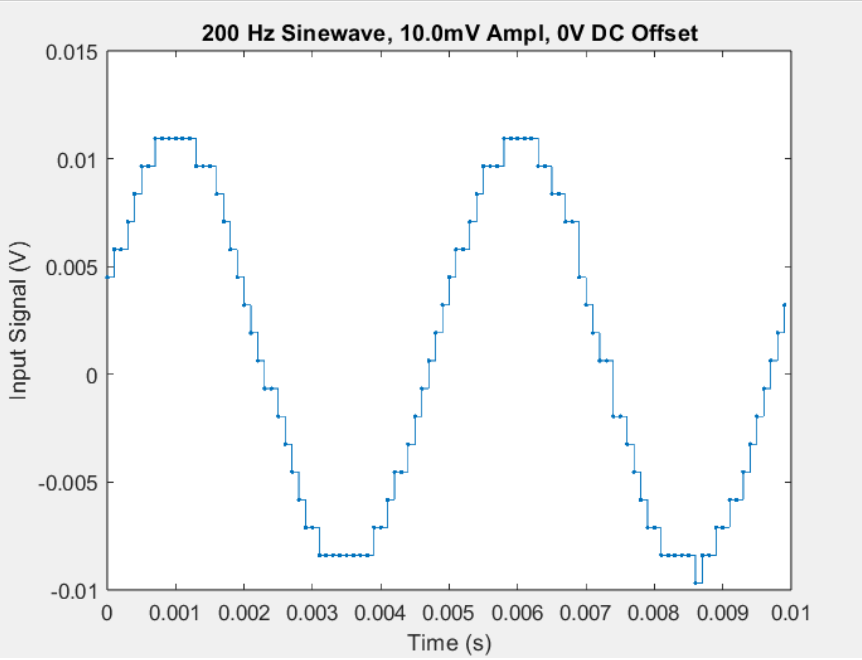


Figure 3: 200 Hz, 10 mV amplitude sine wave acquired at 10,000 samples/s (Configuration 2).

The cause was quantization. The signal was so small compared with the ±10 V range of the ADC that each sample was rounded to one of only a few levels. This could have been avoided by amplifying the signal before the DAQ, or by using an ADC with more bits or a smaller input range so that each level is a smaller voltage.

5. Voltage Resolution of the A/D

The NI USB-6001 is a 14-bit ADC with a −10 to +10 V range, so it has:

$$N = 2^{14} = 16{,}384 \text{ levels}$$

with a step size between levels of:

$$\Delta V = 20 \text{ V}/16{,}384 \approx 1.22 \text{ mV}$$

This explains both results. The 400 mVpp signal in configuration 1 spans about 400/1.22 ≈ 328 levels, so quantization was not noticeable. The 20 mVpp signal in configuration 2 spans only 20/1.22 ≈ 16 levels, which matches the staircase in Fig. 3. The measured step of about 1.3 mV is also close to the predicted 1.22 mV.

Two things in Fig. 3 cannot be explained by quantization. The waveform was centered at about +1.3 mV instead of 0 V, and the second trough had one sample at −9.7 mV that the first did not. These come from a small DC offset and electrical noise, which only show up when the signal is this small.

6. Sampling Rate and Aliasing

A 1000 Hz sine wave with a 4 V amplitude was recorded for 2 s at Fs = 20,000, 10,000, 4000, 2500, 1800, 1600, 1400, and 1200 samples/s, and each recording was played back with soundsc(data, Fs). A sine wave that is sampled too slowly shows up at a lower alias frequency:

$f_{alias} = |f - k \cdot Fs|$, with k chosen to make f_{alias} as small as possible

Above 2000 samples/s, Fs is more than twice 1000 Hz, and the tone sounded unchanged. Below 2000 samples/s, the tone was no longer recognizable, and the recordings at 1800, 1600, 1400, and 1200 samples/s all sounded roughly the same, even though the predicted aliases were 800, 600, 400, and 200 Hz. This was likely because soundsc plays these back at very low sample rates and small speakers do not reproduce low frequencies well.

The test was repeated with a 2000 Hz signal at Fs = 20,000, 10,000, 4000, 3600, 3200, 2800, and 2400 samples/s. The tone sounded correct at 20,000 and 10,000 samples/s, quieter and unstable at exactly 4000 samples/s (2 samples per period), and unrecognizable below that, with the recordings all sounding roughly the same. The cutoff moved from 2000 to 4000 samples/s because doubling the signal frequency doubles the minimum sampling rate.

These results show that the maximum frequency that can be digitized is:

$f_{max} = Fs/2$

which is 10,000 Hz, 2500 Hz, and 500 Hz for Fs = 20,000, 5000, and 1000 Hz. Signals above f_{max} are not lost but fold down to a lower frequency.

7. Sampling Audio Signals

After the tone testing, an audio signal from the computer was adjusted to about 1 to 2 Vpp on the oscilloscope, recorded for a few seconds at 20 kHz, 5 kHz, and 1 kHz, and played back with soundsc.

The 20 kHz recording sounded nearly identical to the original. The 5 kHz recording was recognizable but duller, with the higher-pitched parts missing. The 1 kHz recording was hard to recognize, sounding muffled and buzzy.

This follows from section 6: the three recordings could only represent frequencies up to 10 kHz, 2.5 kHz, and 500 Hz. Since the DAQ has no filter to remove frequencies above Fs/2, those frequencies aliased down instead of being dropped, which distorted the lower-rate recordings.

III. Discussion and Conclusions

This lab showed how an ADC turns a continuous signal into numbers, and how its voltage resolution and sampling rate limit how well the original signal is kept. At 10,000 samples/s, a 500 Hz sine wave was captured cleanly. At 1500 samples/s, a 200 Hz signal had only 7.5 samples per period and turned blocky, and at a 10 mV amplitude, the signal covered only about 16 of the 1.22 mV levels of the ADC and became a staircase.

The tone and audio tests showed that a signal can only be recorded up to Fs/2. The 1000 Hz and 2000 Hz tones became unrecognizable once Fs fell below twice their frequency, and the audio became more distorted at lower sampling rates as more of it aliased. Overall, this lab connected sampling theory from class to real measurements, thus demonstrating two ideas: a signal must be large compared with the step size of the ADC to avoid quantization error, and it must be sampled at more than twice its highest frequency to avoid aliasing.

IV. References

[1] Dr. Iman Salama. “Lab 8 – Getting Started with Analog to Digital Conversion and Sampling.” Northeastern University. 16 October 2023.